Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/09/2006
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Project Title:	A Novel Biodosimetry Method For Monitoring Radiation-Induced Genetic Damage		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	53711-5399	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	06/16/2003	**End Date:**	06/16/2007
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: no cost extension through 6/16/2007, per J. Dardano 2/06		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Performance Goal No.:			
Performance Goal Text:			
	For extended space missions, exposure to hazardous radiation from galactic cosmic radiation and solar particle events poses a serious threat to the health of the astronauts. Dosimeters measure radiation dose received but not its biological effects or individual differences in radiation sensitivity. Accurate methods for measuring the extent of an individuals genetic damage resulting from accumulated radiation exposure is critical for estimating personal health risks and need for countermeasures. The overall goal of this proposal is to investigate the efficacy of a novel biodosimetry method for monitoring mutational load resulting from radiation-induced mutations in repetitive DNA sequences. Repetitive sequences constitute nearly one third of the human genome and some have been shown to be extremely sensitive to radiation-induced mutations. Most repetitive sequences are in non-coding regions of the genome and, therefore, alterations in these loci are usually not deleterious to the survival of the host cell. Thus, mutations in non-coding		

Task Description:	repetitive sequences can accumulate, providing a stable molecular record of DNA damage from all past exposures. Progenitor stem cells are long lived and continually divide producing identical duplicate progeny cells. Mutations in repetitive DNA loci in progenitor stem cells will accumulate and through analysis of progeny blood or sperm cells can serve as biological sensors to monitor radiation-induced genetic damage. To test this hypothesis we will (a) screen a broad range of highly mutable repetitive DNA sequences to identify genetic loci most sensitive to high-LET radiation-induced mutations, (b) show that radiation-induced mutations in progenitor stem cells are cumulative and that mutational load can be quantitatively measured by sampling a population of single or small pools of progeny blood or sperm cells for changes in repetitive DNA sequences, and (c) investigate possible mechanisms causing radiation-induced genomic instability in repetitive DNA loci. Knowledge obtained from completion of this proposal will determine the efficacy of a novel quantitative method for monitoring radiation-induced mutational load that would be applicable to both mouse models and humans. Mutational load profiling, through analysis of changes in repetitive DNA sequences, is a simple, non-invasive and generalized approach for monitoring an individual's cumulative record of radiation-induced mutations that would be useful in predicting and minimizing health risks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Mutational load profiling, through analysis of changes in repetitive DNA sequences, is a simple, non-invasive and generalized approach for monitoring an individual's cumulative record of mutations that would be useful in predicting and minimizing health risks for astronauts or any individual exposed to ionizing radiation or chemical mutagens. Preliminary data suggest that the markers identified in this study are also sensitive biomarkers for oxidative stress and may be useful in early detection of conditions that could lead to cancer and other degenerative diseases and may be useful in the helping to develop effective countermeasures.
Task Progress:	The first goal to screen repeat sequences to identify those most sensitive to radiation-induced mutations has been accomplished. Three classes of microsatellite repeat markers have been identified that are sensitive to radiation-induced mutations in mouse and human cells. These include mononucleotide repeats with polyA runs of over 40bp, tri-, tetra- and penta-nucleotide repeats on the Y chromosome, and some A-rich pentanucleotide repeats. The length of the polyA tract in mononucleotide repeats was found to be positively correlated with sensitivity to radiation. The second goal to test whether radiation-induced mutations in repeats accumulate in dose dependent manner has been partially completely. Selected repeat markers did exhibit a dose dependent response in normal human fibroblasts exposed to 0.5 to 3 Gy iron ions. To test whether this dose-dependent response also occurred in blood, sperm and cheek cells we exposed mice to 0.5 to 3 Gy iron ions at 1 GeV then tissue samples were collected 10 weeks post-irradiation. This is sufficient time to allowed cells directly exposed radiation to be replaced from stem cell progenitors. A statistically significant increase in mutation frequency in mononucleotide repeats was observed in blood, sperm and cheek cells indicating radiation-induced mutations were stable over time in stem cell populations. We are currently testing mice that have received two radiation exposures with 10 week interval after each exposure. This will allow us to determine if radiation-induced mutations are additive and stable over longer periods of time. Mice receiving multiple exposures displayed a dose-dependent acceleration of ageing effects such as premature graying, reduction in sperm counts and tumor development. The third goal to investigate role of DNA mismatch repair and oxidative stress in radiation-induced mutations in repeats has been partially completed. Repeat markers that are sensitive or resistant to radiation-induced mutations responded similarly oxidative stress. For example, mutations in Y-STRs showed a dose response to hydrogen peroxide exposure. Mice exposed to paraquat, which increases super oxide levels, displayed increased frequency of mutations in the same markers that showed radiation sensitivity. We are currently investigating the role of DNA mismatch repair and antioxidant enzymes on radiation sensitivity of repeat markers. Our research indicates that some repeat DNA markers are in fact sensitive to radiation-induced mutations in both mouse and human cells. Selected repeat markers show a dose-dependent response to radiation exposure and are stable in mice for at least 10 weeks. Thus, the use of repeat markers for biodosimetry appears feasible. Interestingly, repeat markers that were sensitive to ionizing radiation were also equally sensitive to oxidative stress. This suggests that reactive oxidative stress plays an important role in radiation-induced mutations in repeat sequences and suggests that antioxidants may be effective countermeasures to reduce DNA damage in coding and non-coding repeats. Additional experiments are underway to better understand how radiation induces mutations in repetitive sequences and to identify an optimal set of markers for development of an effective multiplex assay for biodosimetry.
Bibliography Type:	Description: (Last Updated: 04/16/2019)
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Patents	B0663648. PCT application filed on October 24, 2005. Oct-2005 Bacher J, Halberg R, Kent M. "Methods and Kits for Detecting Mutations."

Patents	B0662481. PCT application filed October 24, 2005. Oct-2005 Kent M, Bacher J, Megid W. "Methods and Kits for Detecting Germ Cell Genomic Instability."